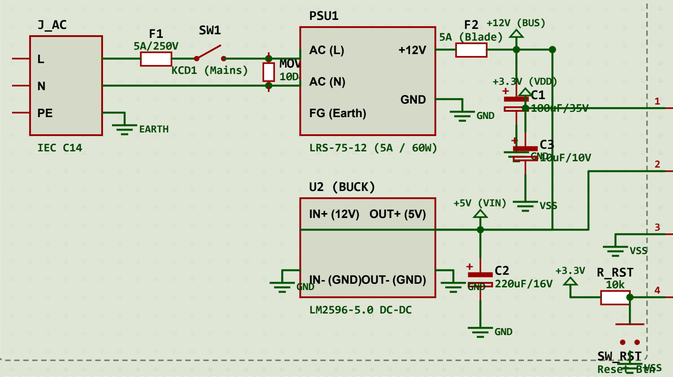
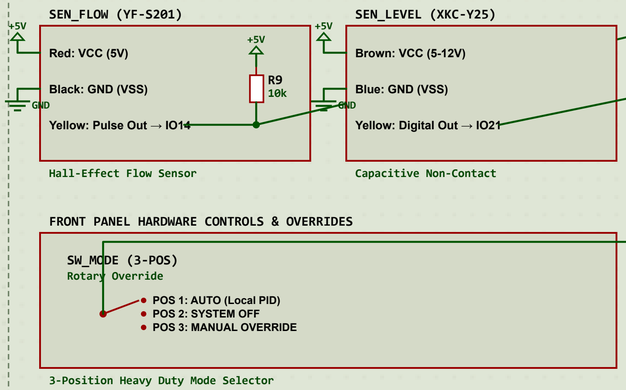
SECTION 1: 220V AC MAINS & DC POWER DISTRIBUTION



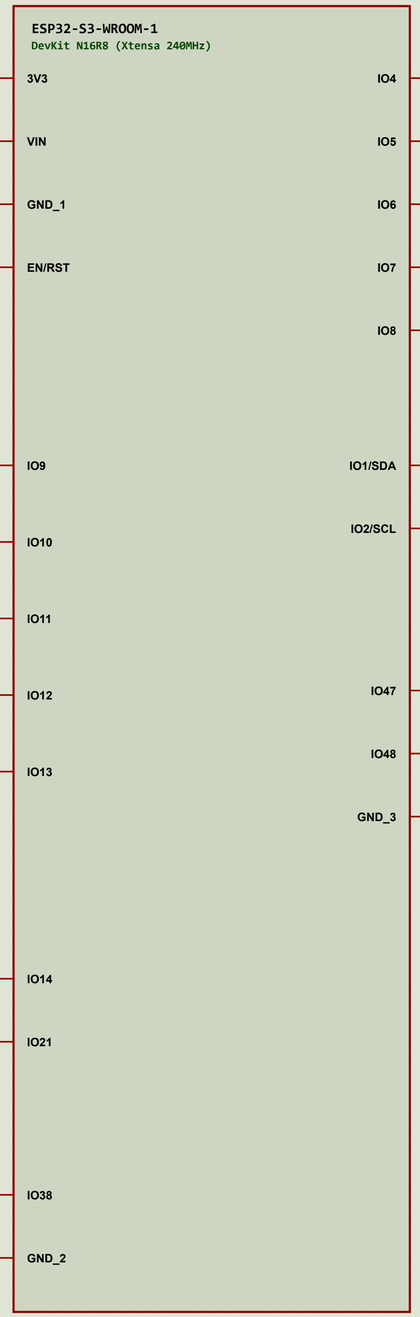
SECTION 2: MULTI-CHANNEL 12V RELAY ACTUATOR CONTROLLERS



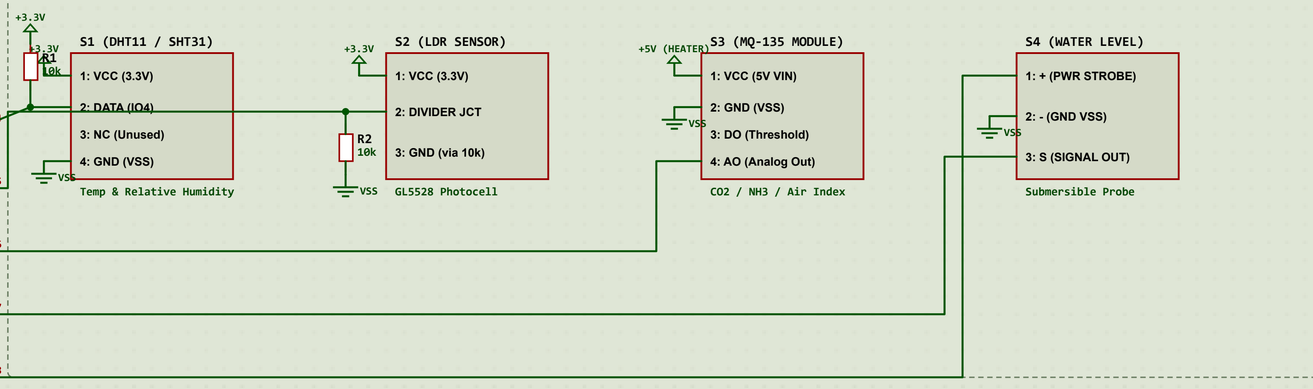
SECTION 3: LIQUID HANDLING SENSORS & HARDWARE CONTROLS



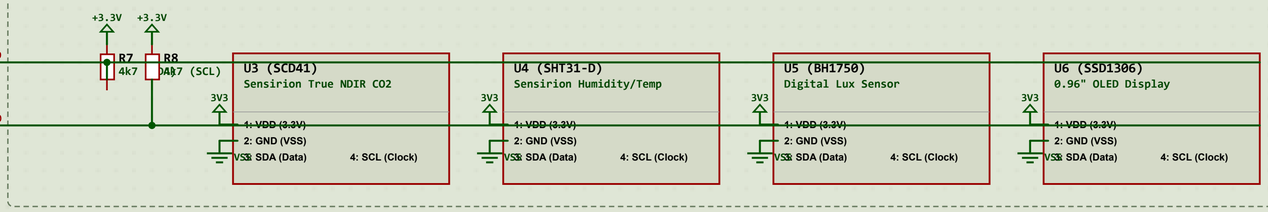
U1



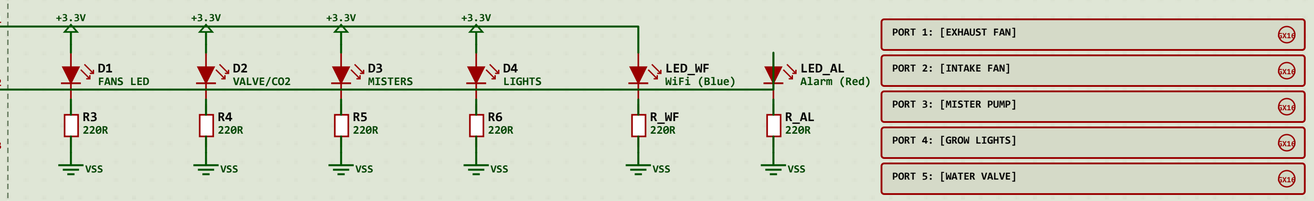
SECTION 4: CORE PROTOTYPE ENVIRONMENTAL SENSORS



SECTION 5: SUSPENDED I2C ENVIRONMENTAL PROBE (WEATHER SHIELD)



SECTION 6: PROTOTYPE LED FEEDBACK & EXTERNAL GX16 PORTS



SECTION 7: COMPLETE PINOUT DIRECTORY & SYSTEM NETLIST

KABUTECH HIYAS — SMART MUSHROOM AUTOMATION SYSTEM

MASTER ELECTRICAL WIRING SCHEMATIC & COMPONENT NETLIST (PROTEUS ISIS SPEC)

DOC ID: KBT-HIYAS-SCH-002
DESIGN: Zandro2005 / Kabutech R&D
MICROCONTROLLER: ESP32-S3 (Xtensa Dual-Core 240MHz)
POWER SPECS: 220VAC In / 12VDC 5A Main / 5VDC / 3.3VDC
REVISION: REV 2.4 (MASTER)
STATUS: BENCH TESTED & VERIFIED

ESP32-S3 COMPLETE PIN ASSIGNMENT REFERENCE

GPIO 4 → DHT11 / SHT31 Data (+10kΩ Pull-up to 3.3V)
GPIO 5 → LDR Light Sensor (10kΩ Divider Junction to GND)
GPIO 6 → MQ-135 Analog Output AO (ADC1 CH5)
GPIO 7 → Water Level Sensor Signal S (ADC1 CH6)
GPIO 8 → Water Level Power V+ (Anti-Electrolysis Gated Strobe)
GPIO 1 → Shared I2C SDA (SCD41, SHT31, BH1750, OLED)
GPIO 2 → Shared I2C SCL (SCD41, SHT31, BH1750, OLED)
GPIO 9 → Relay 1 / D1 LED: Exhaust Fan (12V DC)
GPIO 10 → Relay 2 / D2 LED: Intake Fan / CO2 Valve
GPIO 11 → Relay 3 / D3 LED: Misting Pump (60W 12V)
GPIO 12 → Relay 4 / D4 LED: Grow Light Strip (12V)
GPIO 13 → Relay 5 / D5 LED: Solenoid Water Valve (12V NC)
GPIO 14 → Water Flow Sensor Pulse In (YF-S201 Hall)
GPIO 21 → Capacitive Water Level In (XKC-Y25 Digital)
GPIO 47 → WiFi Status LED (Blue)
GPIO 48 → System Alarm LED (Red)
GPIO 38 → Rotary Mode Selector (Auto/Man)
GPIO 0 → Setup / Provisioning Button
EN / RST → Hardware Reset Button